Name of Student : Raghav Mahesh Sarda	**PRN:** 25070122***235***
Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 5	**Year: AY 2026-27**
TITLE : Collection Classes and StringBuffer	

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Program:

ToDoList.java

```
// Assignment 5 - Exercise 1: To-Do List using ArrayList and StringBuffer
import java.util.*;
public class ToDoList
{
   public static void main(String args[]) // Main method
   {
       Scanner sc = new Scanner(System.in);
        ArrayList<String> tasks = new ArrayList<String>(); // ArrayList for dynamic task storage

       System.out.print("Enter number of tasks: ");
       int n = sc.nextInt();
       sc.nextLine(); // Consuming leftover newline

       for(int i=0; i<n; i++) // Loop to accept tasks
       {
           System.out.print("Enter Task " + (i+1) + ": ");
           tasks.add(sc.nextLine()); // Adding user input to ArrayList
       }
```

```
        StringBuffer sb = new StringBuffer(); // StringBuffer for string
manipulation
        sb.append("\n- TO-DO LIST -\n"); // Appending header

        for(int i=0; i<tasks.size(); i++) // Iterating through ArrayList
        {
            sb.append((i+1) + ". " + tasks.get(i) + "\n");
        }

        System.out.println(sb.toString());
        sc.close();
    }
}
```

StudentCourseRegistration.java

```
// Assignment 5 - Exercise 2: Course Registration using ArrayList and
StringBuffer
import java.util.*;

public class StudentCourseRegistration
{
    public static void main(String args[]) // Main method
    {
        Scanner sc = new Scanner(System.in);
        ArrayList<String> courses = new ArrayList<String>(); // ArrayList
to store courses dynamically

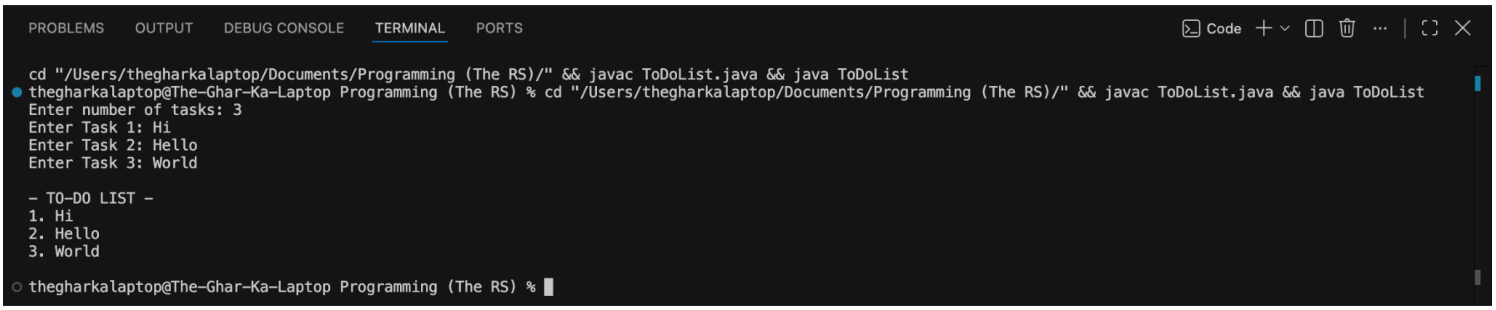
        while(true) // Infinite loop for menu
        {
                System.out.println("\n1. Add Course  2. Remove Course  3.
View Courses  4. Exit");
            System.out.print("Enter choice: ");
            int ch = sc.nextInt();
            sc.nextLine(); // Consuming leftover newline

            if(ch == 1) // Add condition
            {
                System.out.print("Enter Course Name to Add: ");
                courses.add(sc.nextLine());
            }
            else if(ch == 2) // Remove condition
            {
```

```
                System.out.print("Enter Course Name to Remove: ");
                courses.remove(sc.nextLine());
            }
            else if(ch == 3) // View condition
            {
                StringBuffer sb = new StringBuffer();
                sb.append("\n=== Registered Courses ===\n"); // Appending
header
                for(int i=0; i<courses.size(); i++)
                {
                    sb.append("- " + courses.get(i) + "\n"); // Appending
course names
                }
                System.out.println(sb.toString());
            }
            else if(ch == 4)
            {
                break;
            }
        }
        sc.close();
    }
}
```

Output:

ToDoList.java



StudentCourseRegistration.java

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                                    Code

cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac StudentCourseRegistration.java && java StudentCourseRegistration
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) % cd "/Users/thegharkalaptop/Documents/Programming (The RS)/" && javac StudentCourseRegistration.java && j
ava StudentCourseRegistration

1. Add Course  2. Remove Course  3. View Courses  4. Exit
Enter choice: 3

=== Registered Courses ===


1. Add Course  2. Remove Course  3. View Courses  4. Exit
Enter choice: 1
Enter Course Name to Add: JAVA

1. Add Course  2. Remove Course  3. View Courses  4. Exit
Enter choice: 1
Enter Course Name to Add: DBMS

1. Add Course  2. Remove Course  3. View Courses  4. Exit
Enter choice: 1
Enter Course Name to Add: DS

1. Add Course  2. Remove Course  3. View Courses  4. Exit
Enter choice: 3

=== Registered Courses ===
- JAVA
- DBMS
- DS


1. Add Course  2. Remove Course  3. View Courses  4. Exit
Enter choice: 2
Enter Course Name to Remove: DBMS

1. Add Course  2. Remove Course  3. View Courses  4. Exit
Enter choice: 3

=== Registered Courses ===
- JAVA
- DS


1. Add Course  2. Remove Course  3. View Courses  4. Exit
Enter choice: 5

1. Add Course  2. Remove Course  3. View Courses  4. Exit
Enter choice: 4
thegharkalaptop@The-Ghar-Ka-Laptop Programming (The RS) %
```

Conclusion:

In this assignment, concepts of dynamic memory management and string manipulation using ArrayList and StringBuffer were successfully implemented and verified. ArrayList provided flexible data structures for storing objects dynamically without fixed size limitations, and StringBuffer enabled efficient string concatenation and display formatting without creating unnecessary duplicate objects in memory.